

Drug Use & Personality Analysis:

Exploring the Link Between Personality Traits and Substance Use - Hannes Jaakson & Richard Miikael Jaks

The dataset is from Kaggle. It contains demographic variables (**age, gender, education, country, ethnicity**) and detailed **drug-use histories** across six levels of use, ranging **from never used to used in the last day**. It also includes personality measures such as the **Big Five traits, impulsivity, and sensation seeking**, (table above) which help explain individual differences in behavior. Data was cleaned by removing all countries other than US and UK, removed all rows with drug “semeron”, since it was a made up drug to get over-achivers. Left school at 16 years includes now everyone who left school before 16 years and the same is with 16-18 years. <https://www.kaggle.com/datasets/mexwell/drug-consumption-classification/data>

The motivation for this project comes from both of the authors interest in data driven information about narcotics users.

- Goal 1: Data mining and cleaning – prepare and preprocess the dataset for analysis.
- Goal 2: Analyze relationships between personality traits, demographics, and drug use; explore links between different substances and cluster individuals by behavioral similarity.
- Goal 3: Visualize results and draw conclusions about patterns and predictors of drug use.

Patterns show a clear divide between universally used substances (alcohol, caffeine, chocolate) and niche or high-risk drugs, whose users cluster into distinct behavioral groups. The near-absence of recent use for substances like VSA and Amyl („Poppers“) likely reflects strong social desirability bias or underreporting rather than true scarcity.

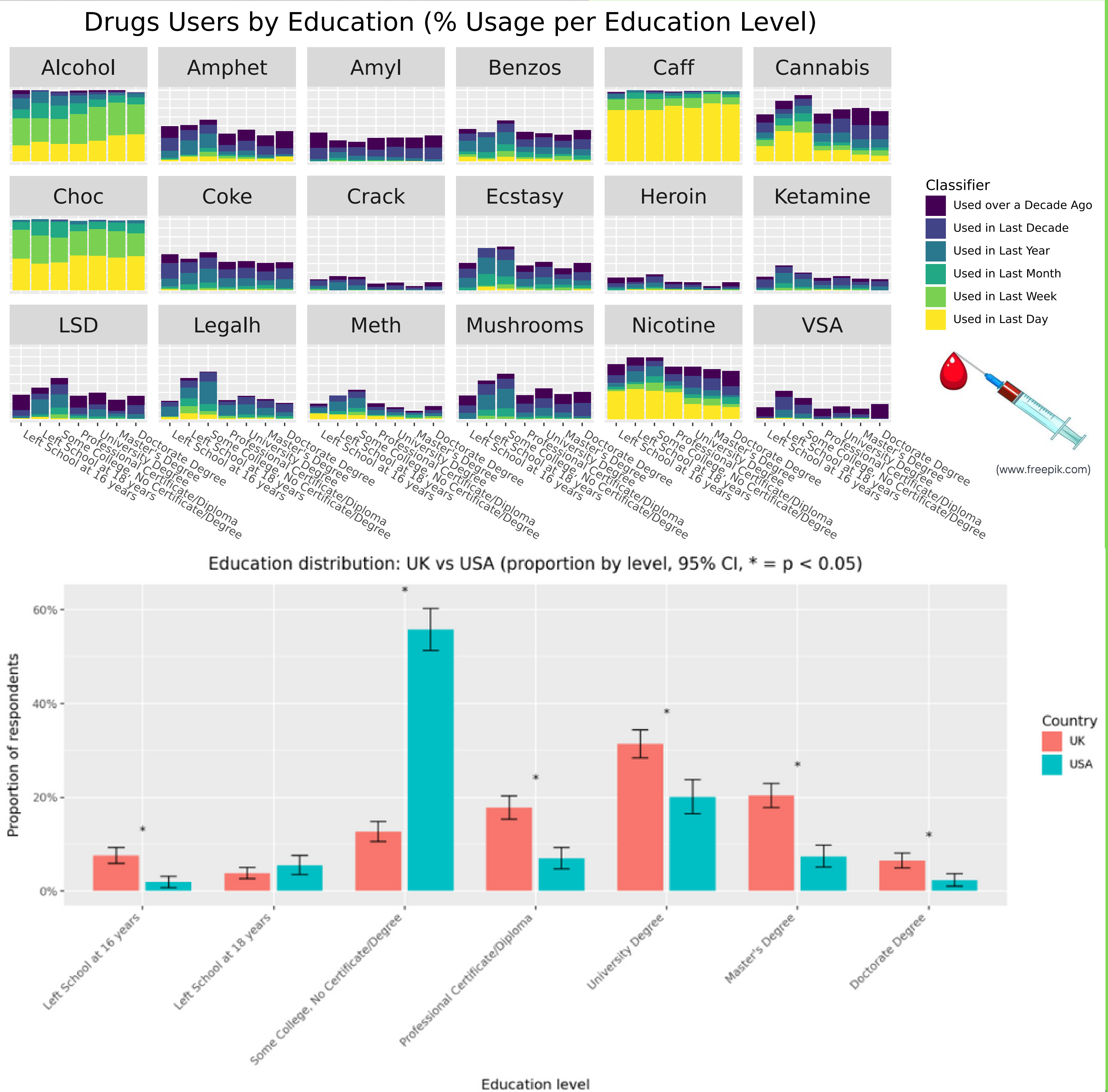
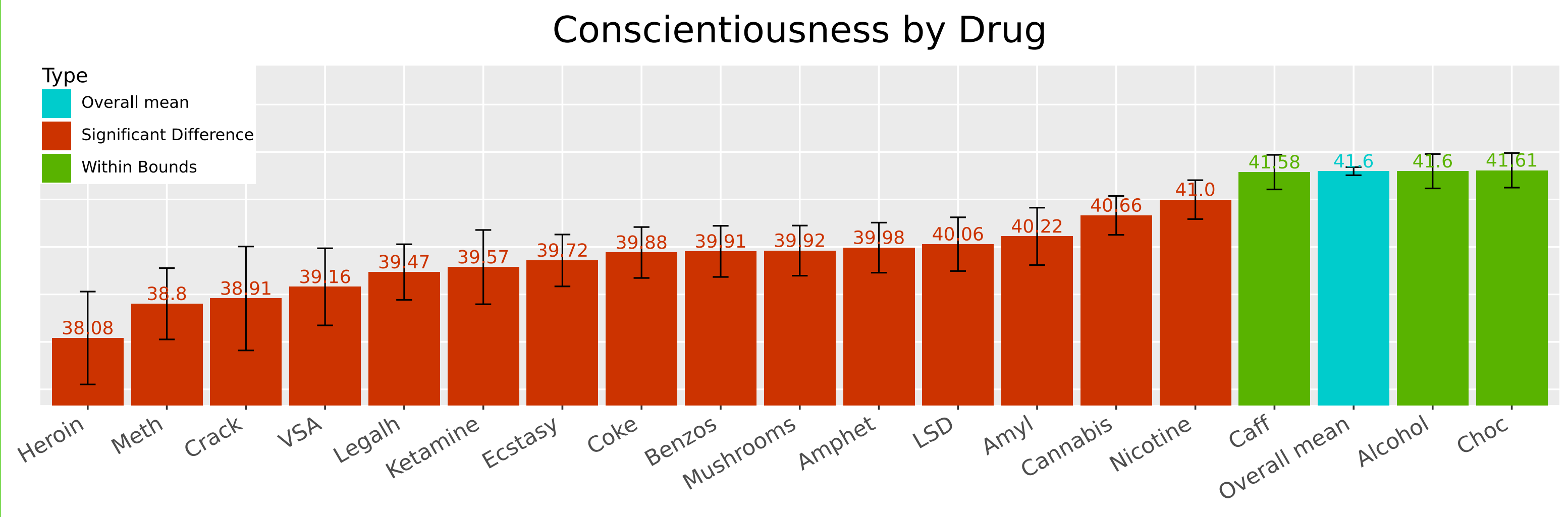
Trait	Explanation	High Scores	Low Scores
Neuroticism	Emotional sensitivity	More stress, mood shifts	Calm, stable
Extraversion	Sociability level	Energetic, outgoing	Quiet, reserved
Openness	Curiosity, creativity	Imaginative, open	Prefers routine
Agreeableness	Cooperation style	Friendly, trusting	Tough, skeptical
Conscientiousness	Self-discipline	Organized, reliable	Flexible, less structured
Impulsivity	Acting without thinking	Spontaneous	Careful
Sensation	Response to sensory input	Likes strong stimulation	Prefers calm

KEY INSIGHTS

- Women use less drugs, especially hard drugs
- **College dropouts do the most drugs**, especially legal highs, nicotine and cannabis
- Young people drink more often, but everyone drinks about the same every month.
- **Older people smoke nicotine more**, while younger addicts prefer cannabis more.
- Young people in general do more drugs.
- In general, drug users almost always had significantly different personality traits.

Comparison to mean of all drugs. p = 0.05

- Users of Heroin, Meth, VSA, Benzos are significantly less extraverted
- Users of Mushrooms, Ecstasy, Amphetamines, Coke, Ketamine, legal highs, volatile substance abuse, crack, meth and heroin were significantly more neurotic.
- Only users of Poppers, caffeine, alcohol and chocolate were NOT significantly less agreeable.
- Only users of caffeine alcohol and chocolate were NOT significantly less Conscientious.
- Only users of alcohol, choc, caff and poppers were NOT significantly more open.
- Only users of chocolate, alcohol and caffeine were NOT significantly more impulsive.



Pairwise clusters Kmeans neighbours = 10

Amphet + Legalh (bath salts, herbal incense) + Ecstasy = “club drug” cluster

Caffeine = universal baseline

Alcohol = universal mixer

Cannabis = unique behavioral profile

Ketamine = between hard drug and party drug

Nicotine = broad-use background substance

Meth + Coke + Mushrooms + LSD = strong co-usage cluster

Heroin + Crack + Amyl (Poppers) + VSA (glues, aerosols) = rare / high-risk pattern cluster

Benzos (Xanax, Valium, Klonopin) = unique sedative profile

